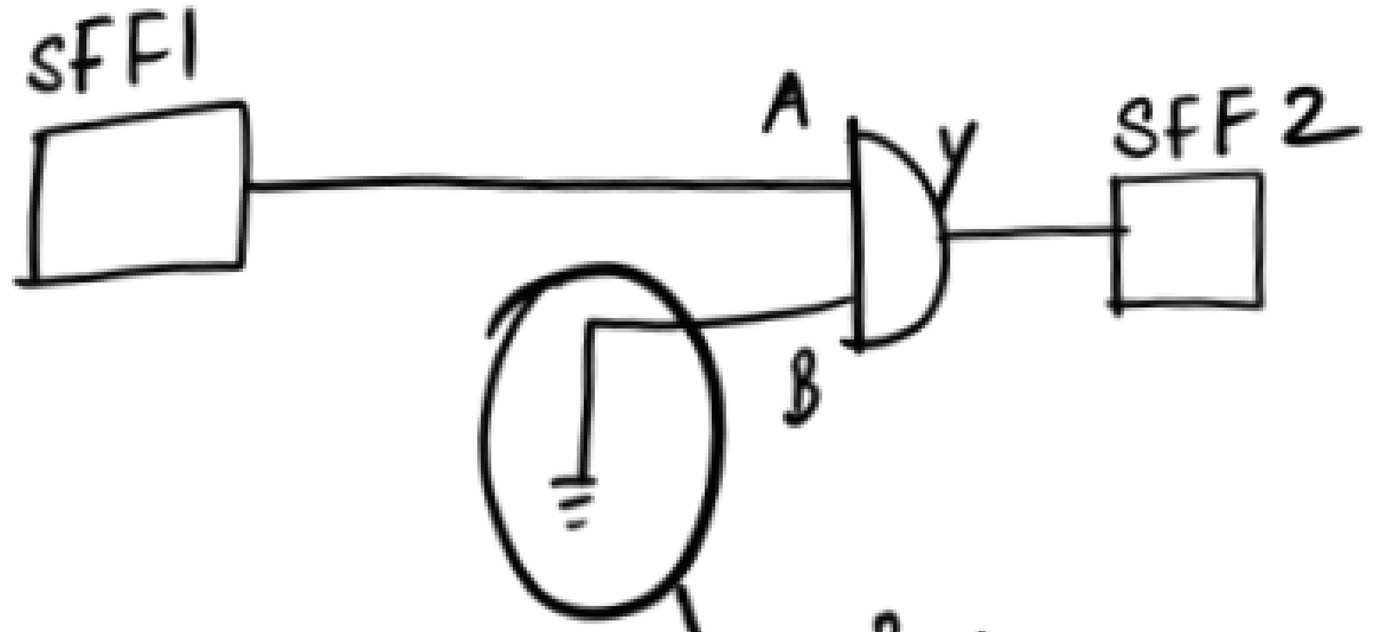
Q:



intentionally done by the designer

How many TDF faults will be detected ?

Ans:- In this case, the o/p of AND gate is always going to be 0. (Since B is tied to Gnd)

∴ Transition can't be created in Y net.

As B net is tied to Gnd, transition can't be created in B net.

In A net, transition can be created but it can't be propagated to an obs pt (SFF2)
[Since B net is tied to ground, Y is always 0]

∴ in this scenario, none of the TDF faults will get detected.

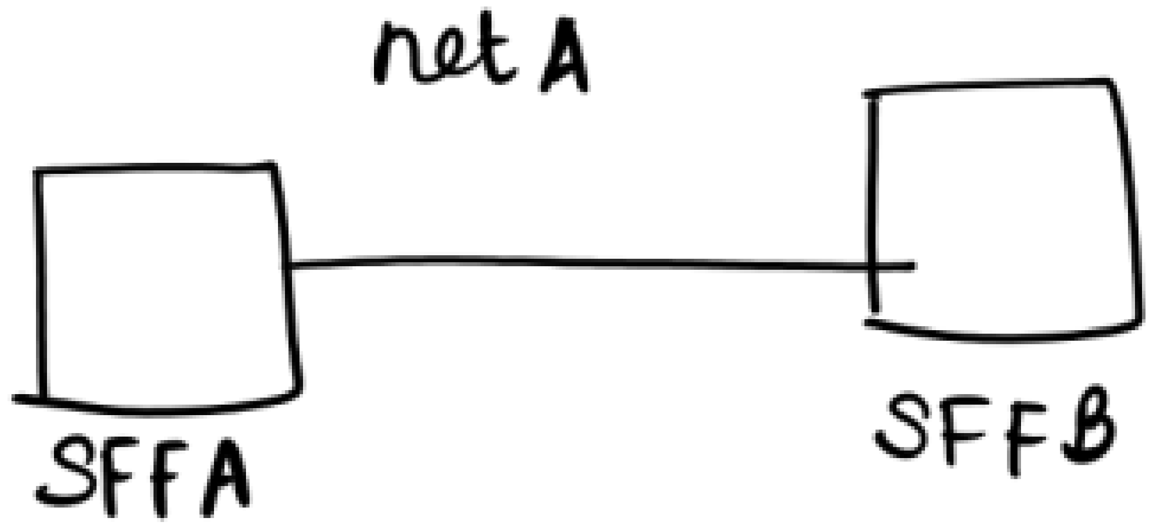
NOTE: During ATPG, tool doesn't consider anything about the delays of cells and nets.

During ATPG, tool will just model/mimic a failure situation (Eg: A str)

Then it will create a transition and it will propagate it to an obs pt.

If it is able to do so, then it is assured that a manufacturing defect with that type of behaviour can be found with that specific test pattern.

Eg:-



The tool will model/mimic a faulty beh

(Eg: A slow to rise)

Then it will try to create a 0-1 transition in net A

and it will propagate it to an obs pt (SFFB)

(Expected value) Good M/C :- value in SFFB (obs pt) after c pulse = 1.

Faulty M/C :- Tool will assume that A is slow to rise
value in the Obs pt (SFFB) after c pulse = 0

After manufacturing, due to the presence of a defect, net A delay is higher than the good net delay.

Eg:- Good net delay = 1ns

Net delay with manufacturing defect = 3ns

T_P of FCLK = 2.5ns

The signal will take more than 2.5ns to rise because of the defect.

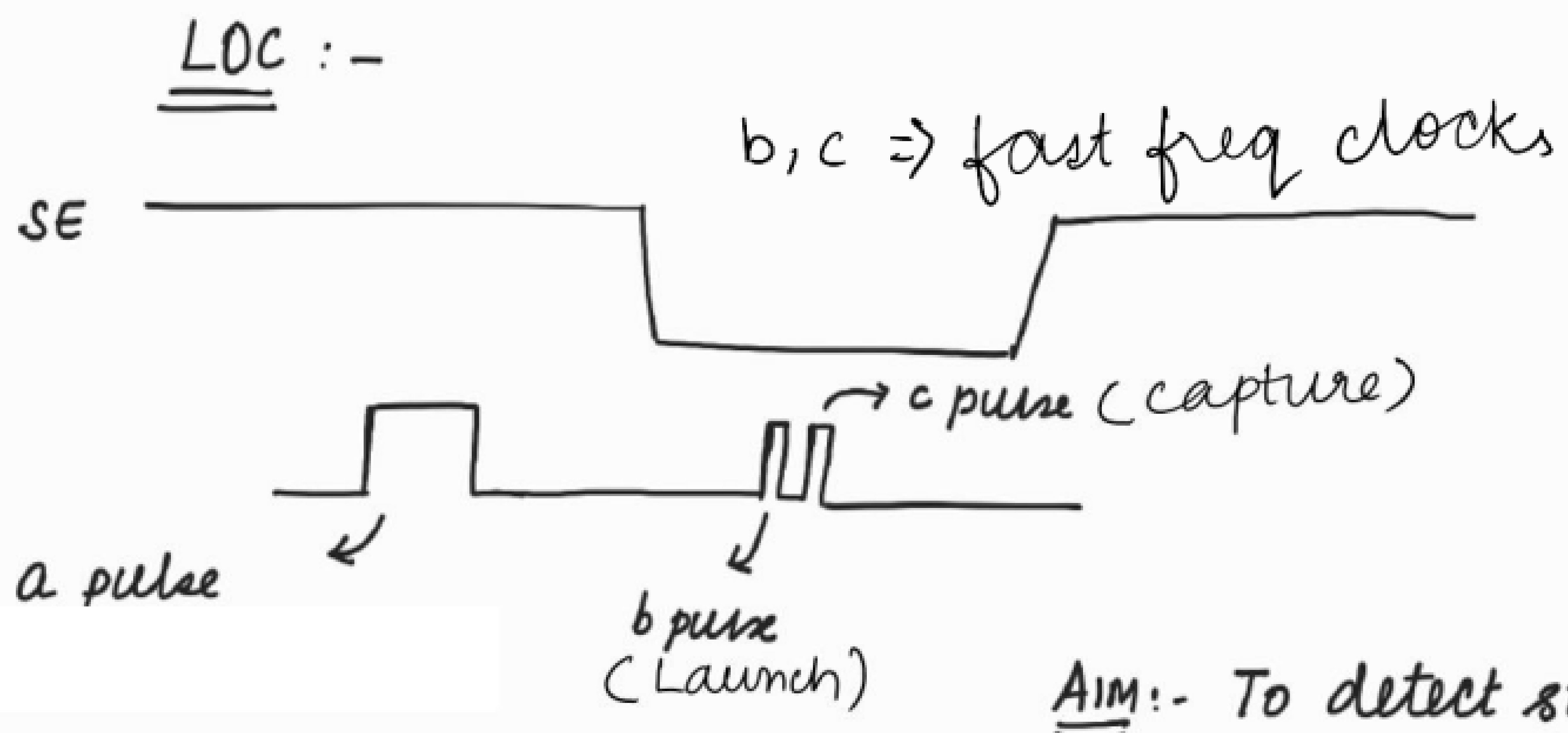
It can be detected with that particular test pattern.

WAYS TO DETECT TDF :

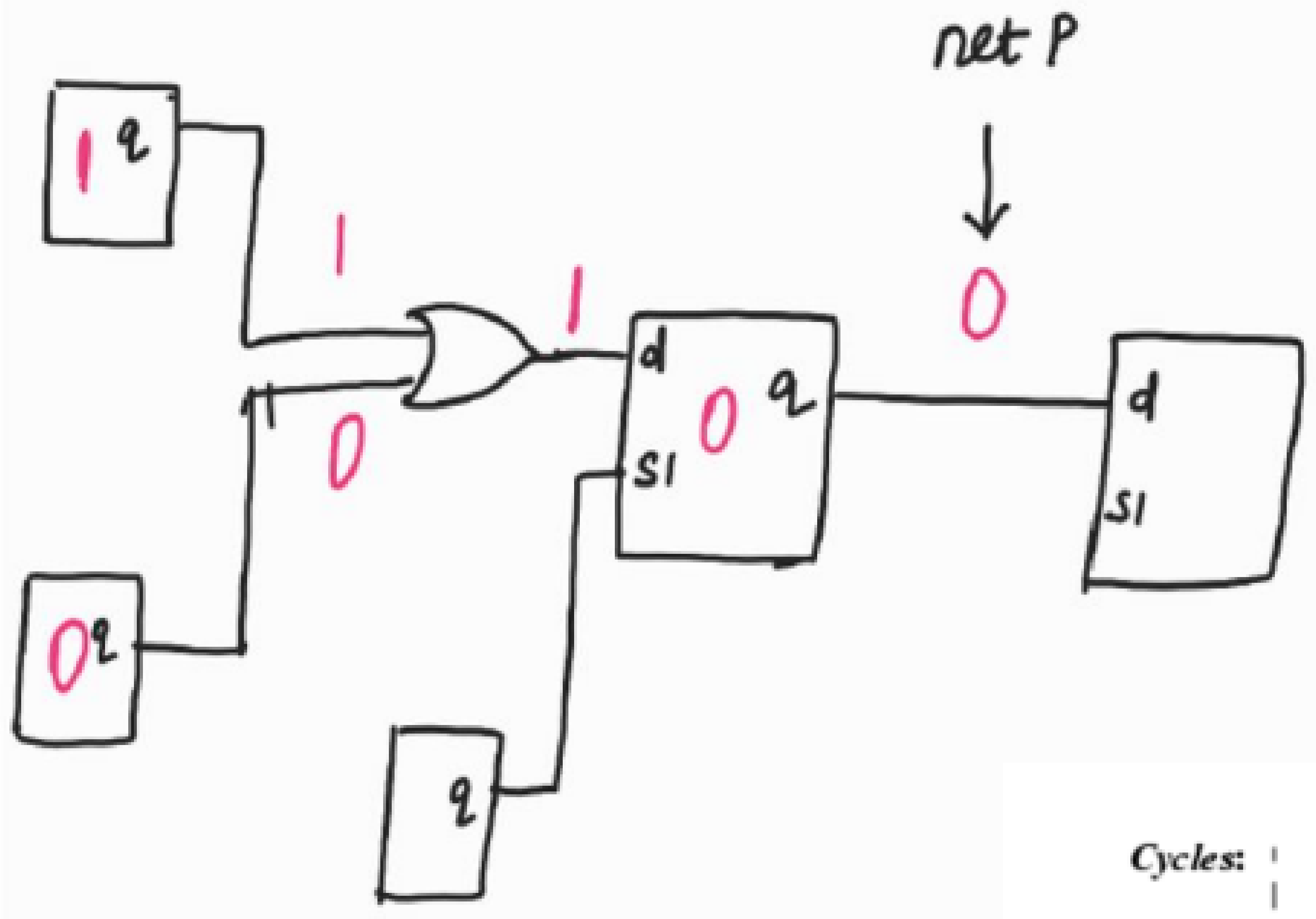
(i) LOC : Launch on capture
Launch off capture

(ii) LOS : Launch on shift
Launch off shift.

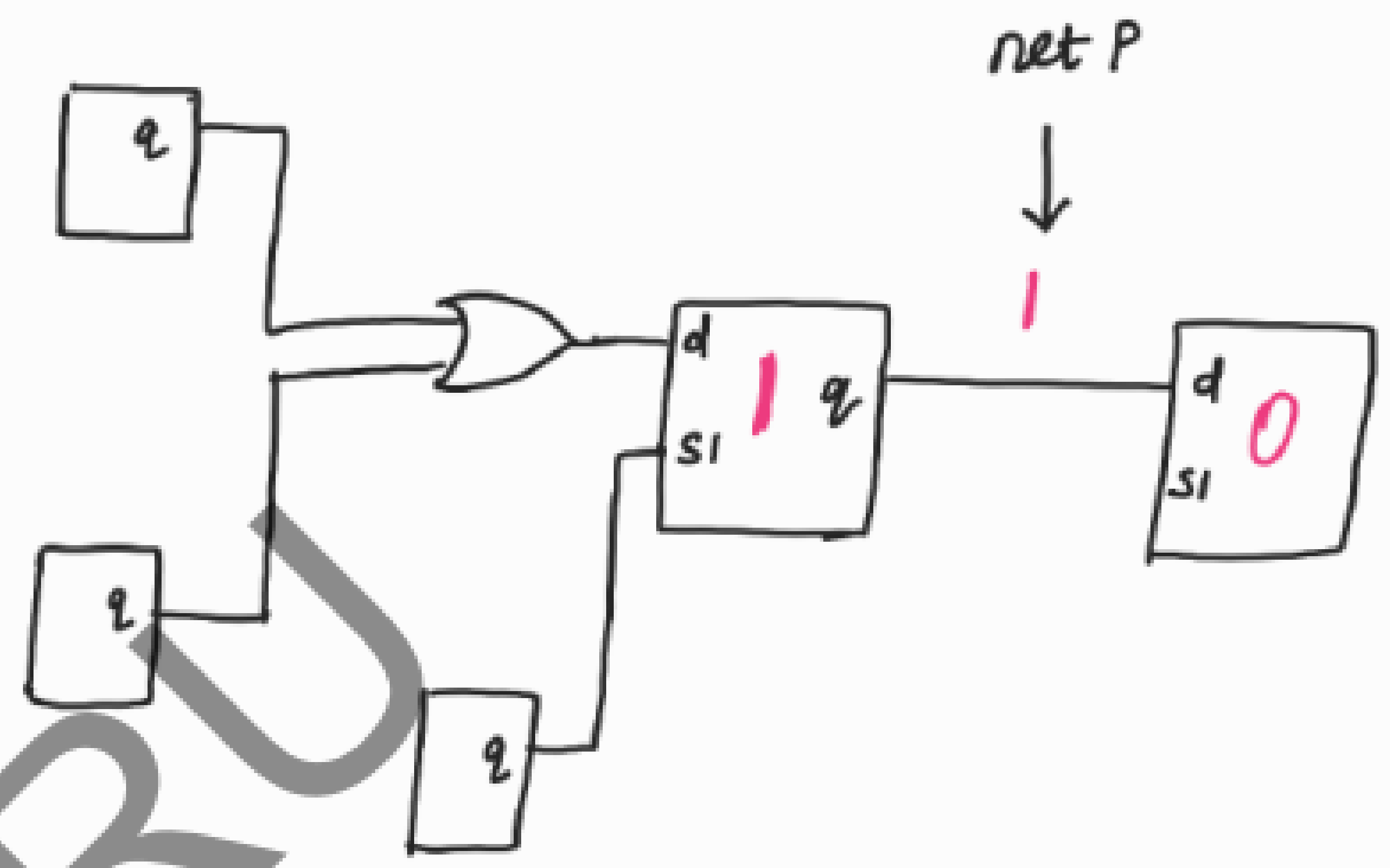
WAYS TO DETECT TDF (Contd…):



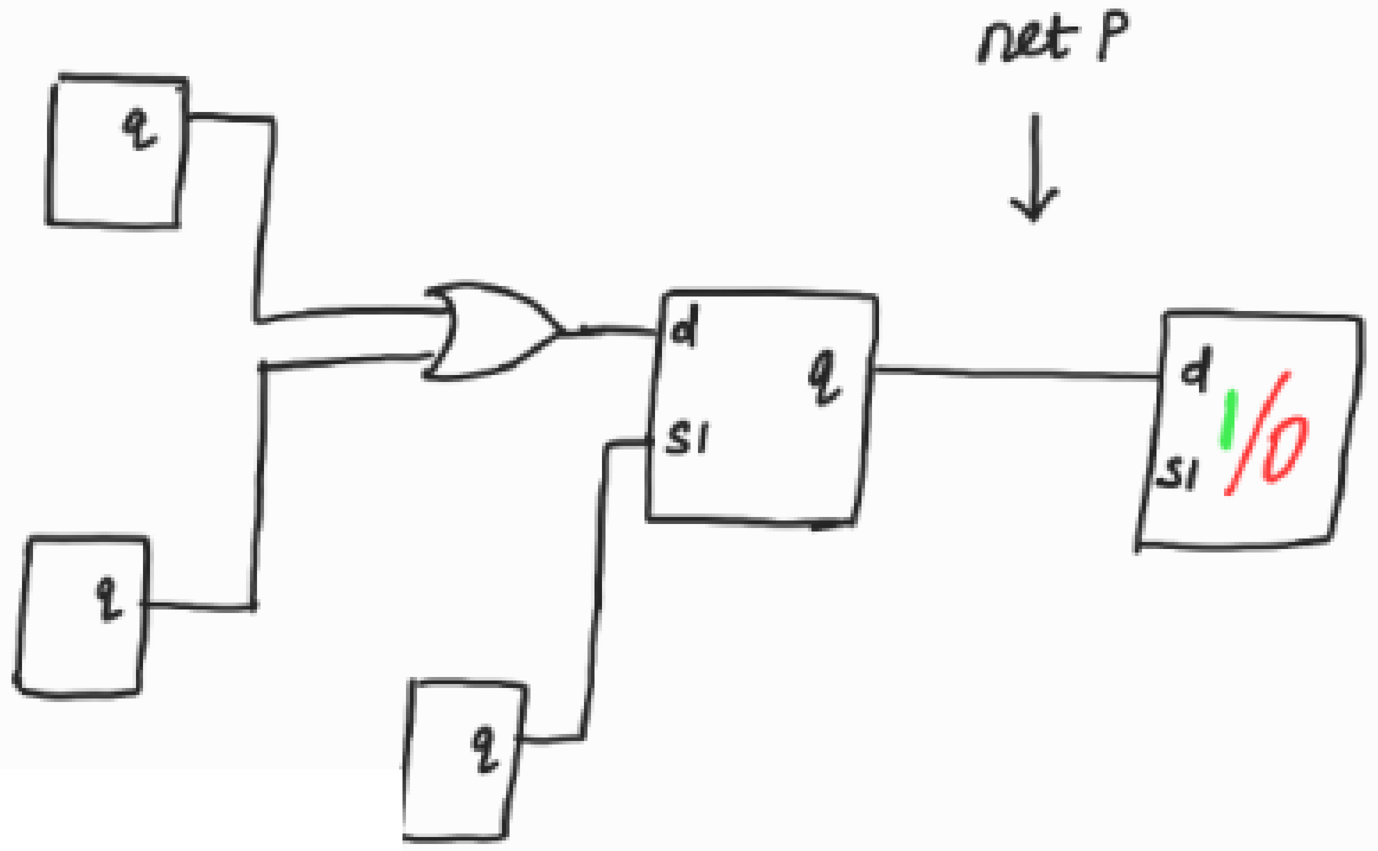
AIM:- To detect str fault in net P.



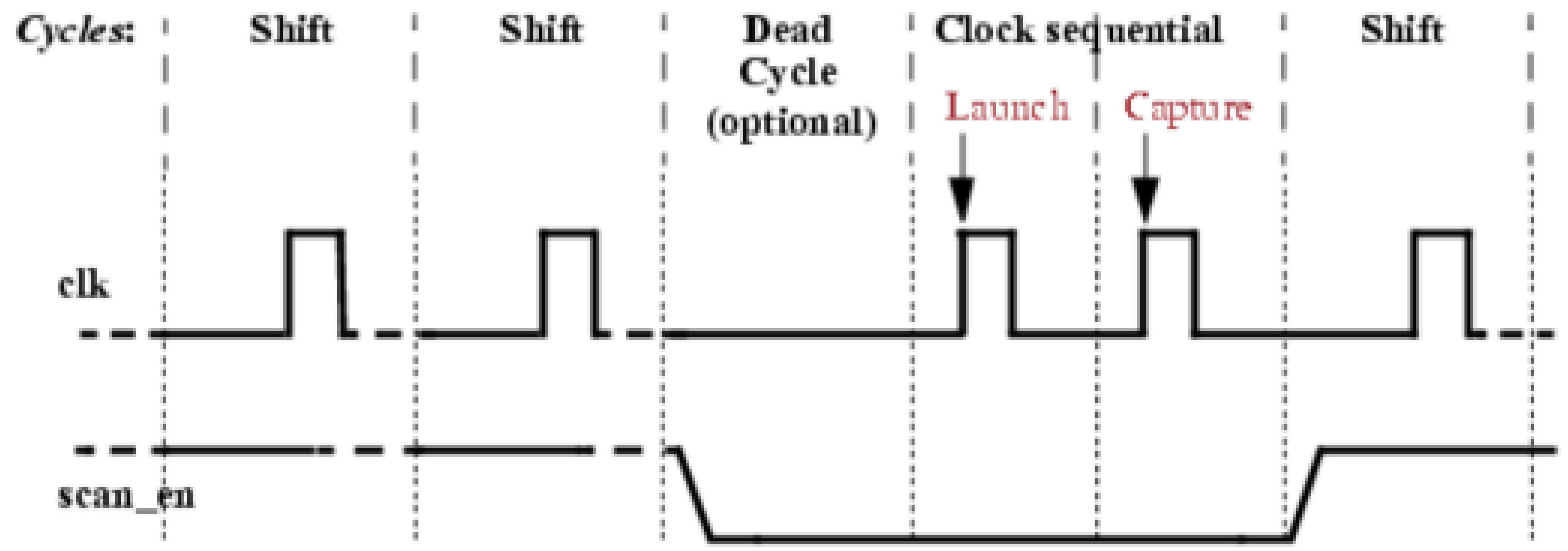
a pulse



b pulse

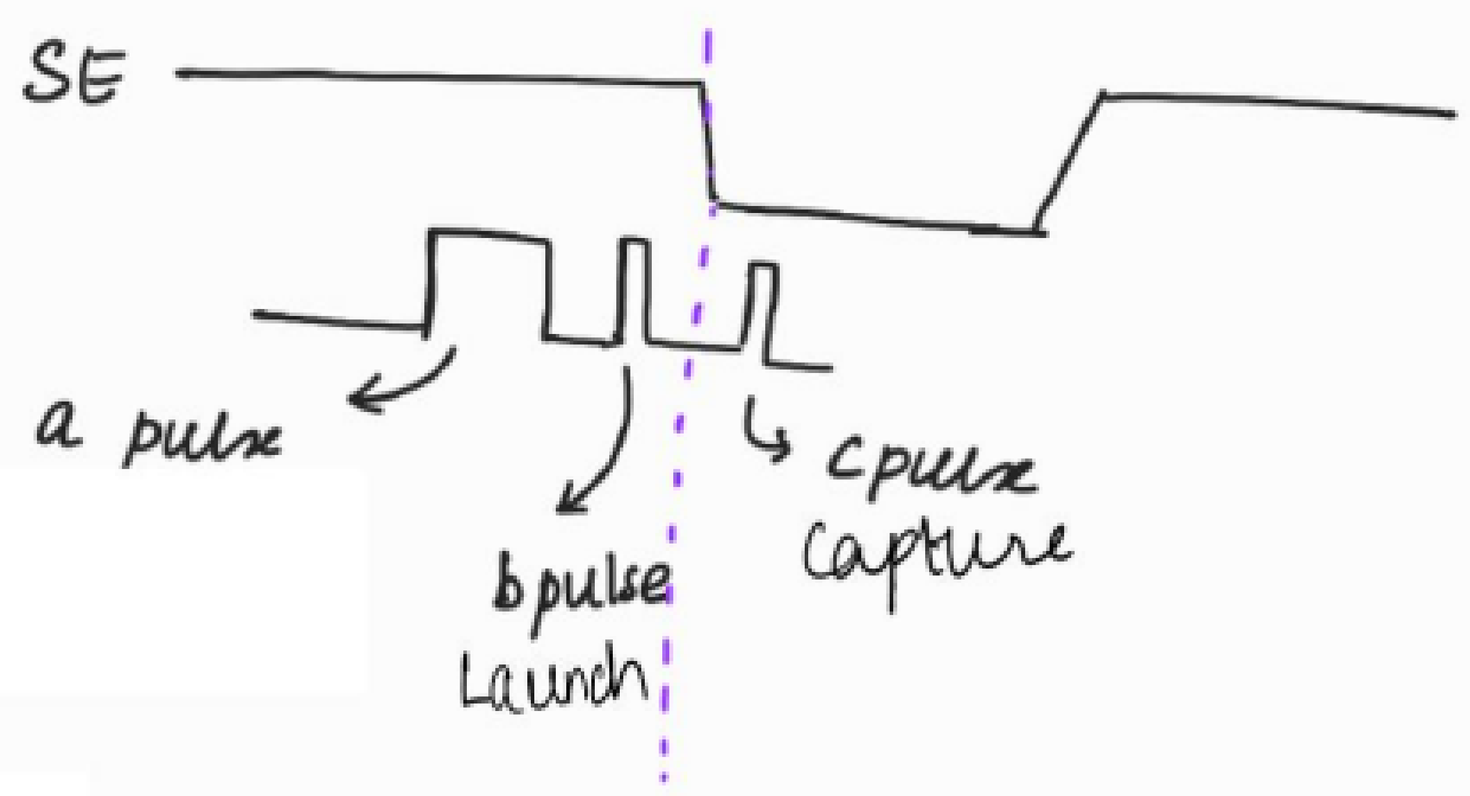


c pulse



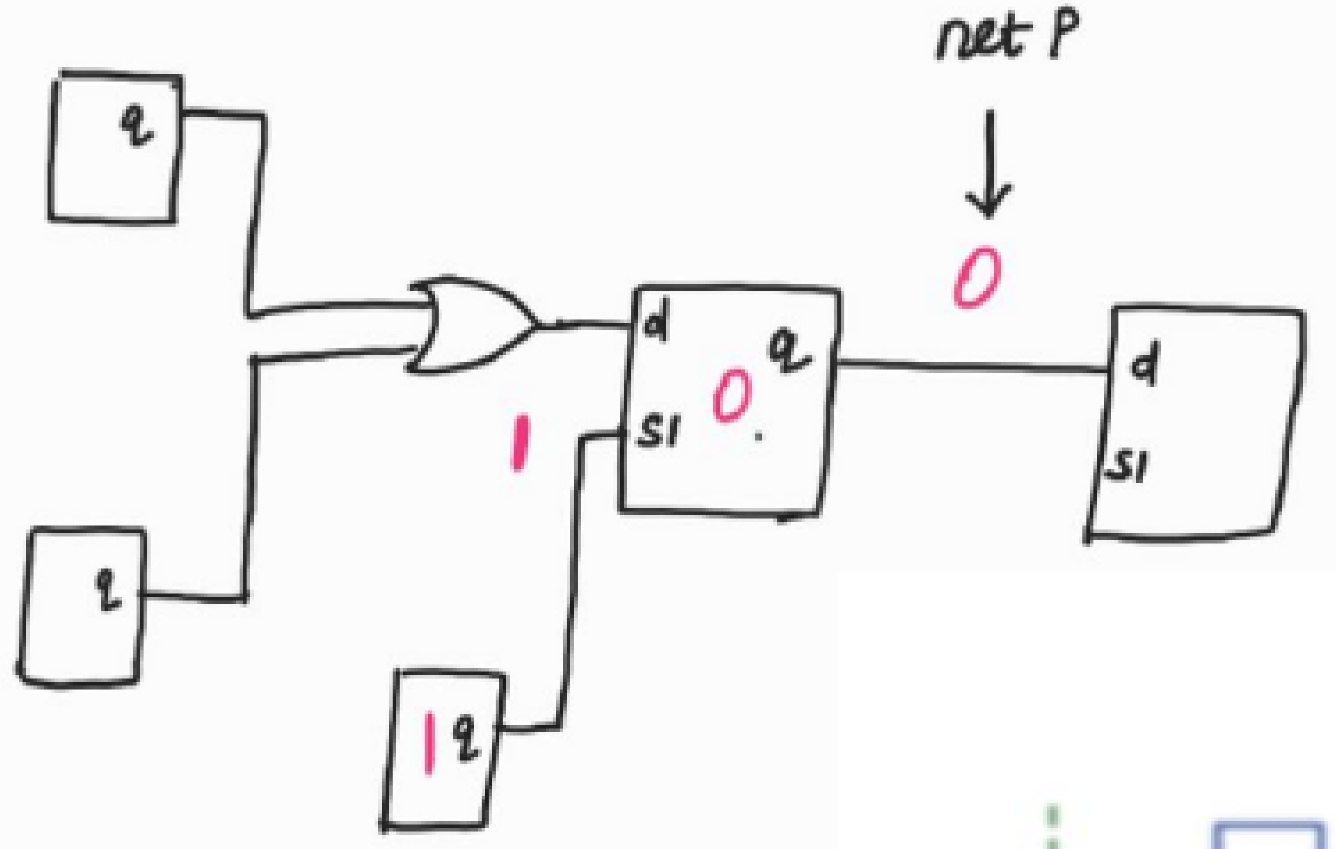
WAYS TO DETECT TDF (Contd...):

LOS :

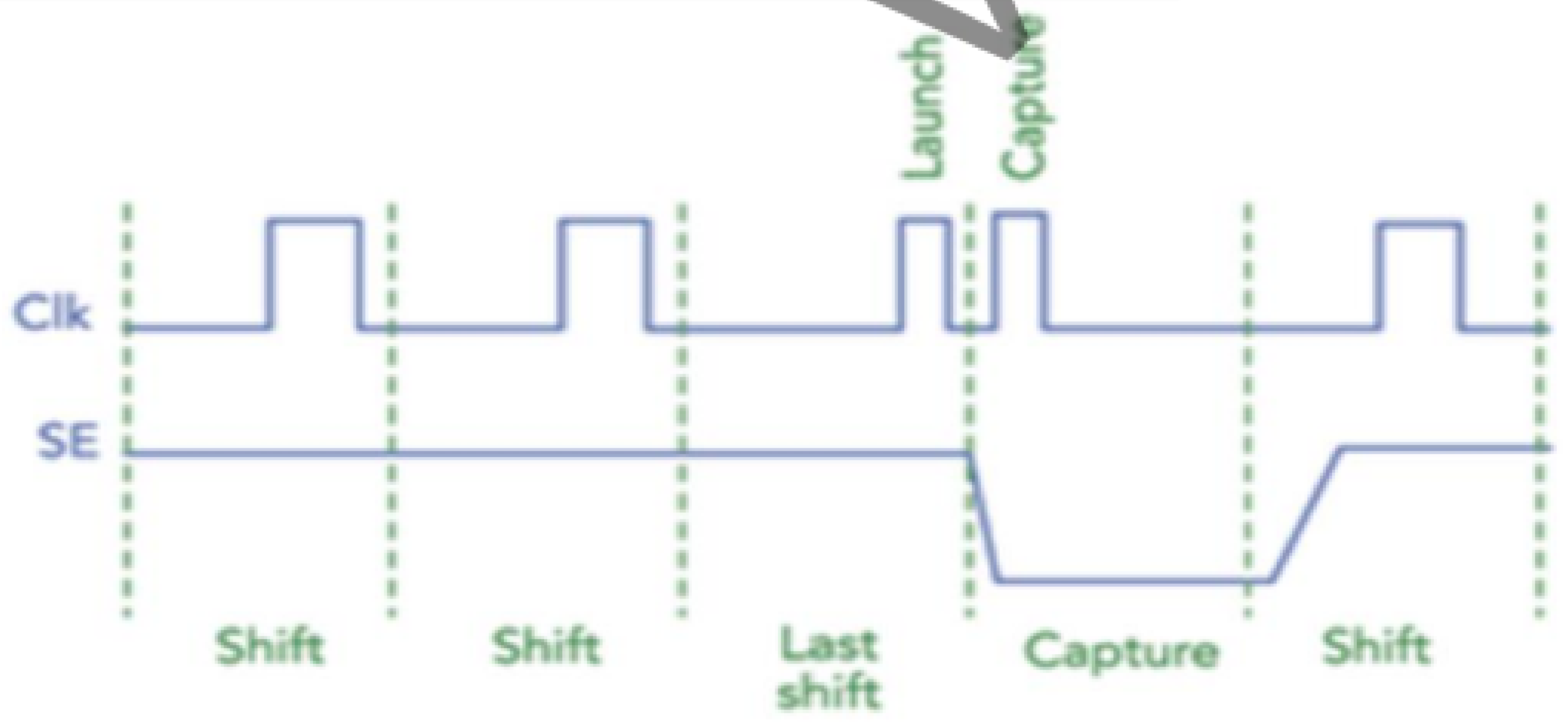


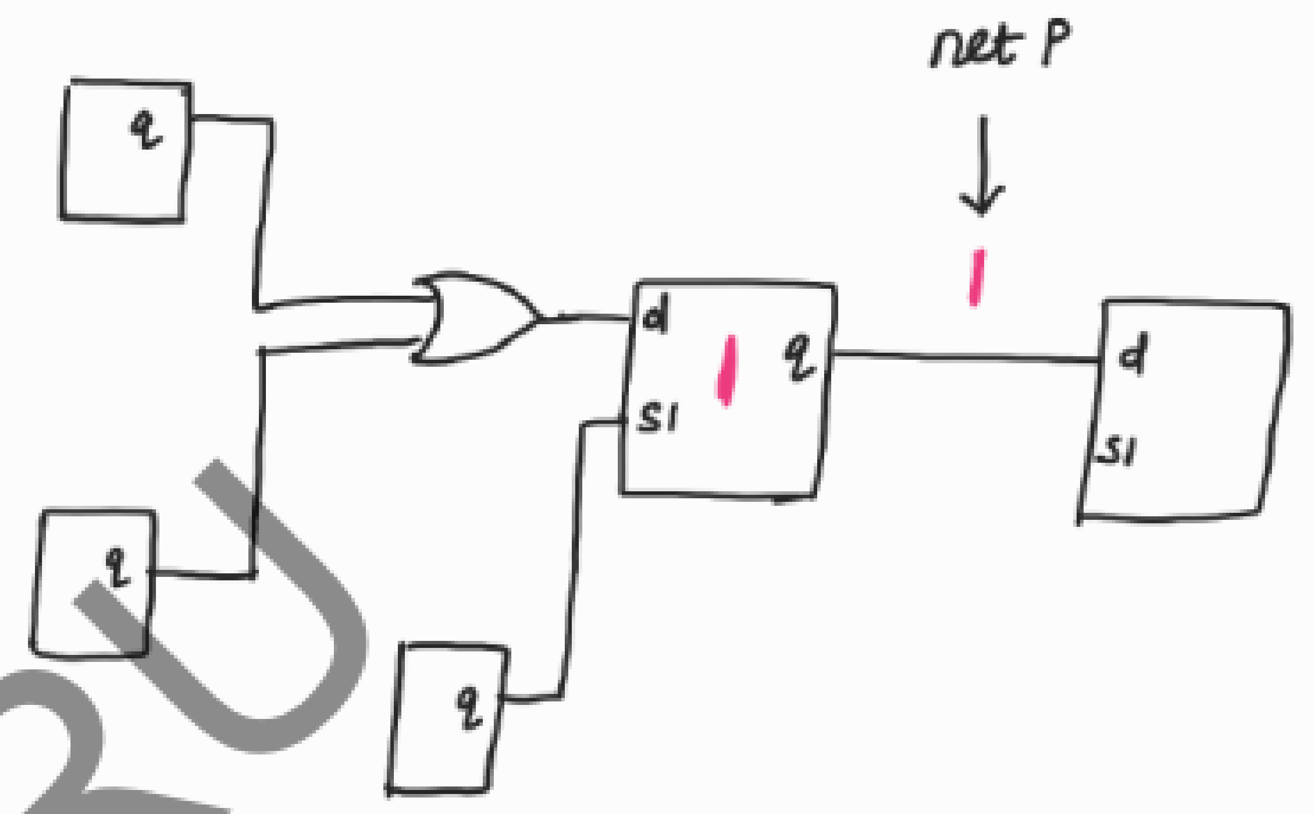
b, c ⇒ fast freq clock

Aim:- to detect str fault in net P.

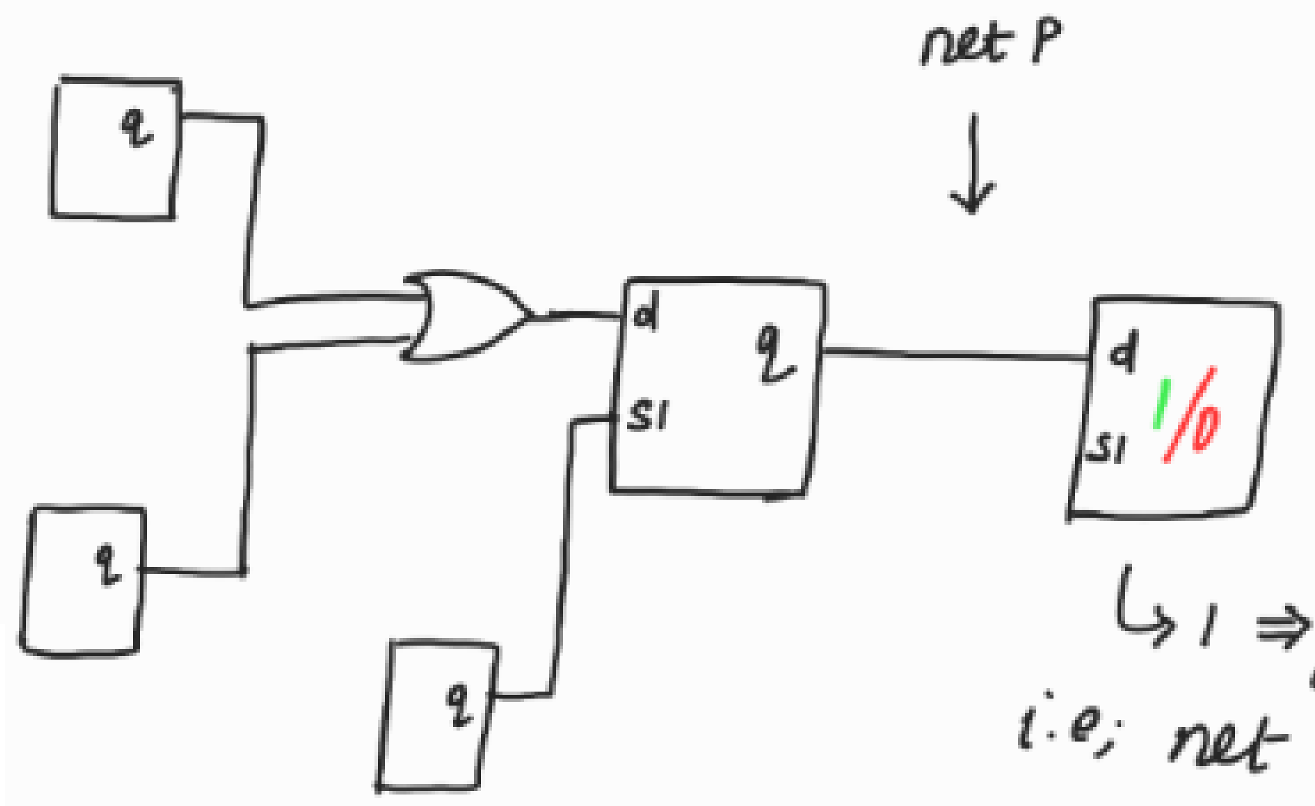


a pulse





b pulse



c pulse

↳ 1 ⇒ good value

i.e; net P is not having str fault.

0 ⇒ faulty value

i.e; net P is having str fault

WAYS TO DETECT TDF (Contd…):

OBSERVATIONS

- LOS has slightly more coverage compared to LOC as the tool has more controllability on LOS than LOC.

Explanation:

The tool brings the transition value through shift path in LOS.
Whereas in LOC, the tool has to program the combo logic to get an appropriate value at the output of the combo logic to create the transition. If suppose the tool is not able to bring the appropriate value at the output of combo logic, then the faults that require that value will not get detected.

WAYS TO DETECT TDF (Contd…):

OBSERVATIONS (Contd…)

- But there is a major limitation in LOS. Controlling SE (timing closure on SE) is extremely difficult.

  Therefore mostly companies don’t use LOS

PIPELINED SCAN ENABLE

In OCC, when SEN=1, we propagate slow frequency clock for both SA and TDF.
How while using LOS method, we are able to propagate fast freq clock (b pulse) when SEN=1?

→ using Pipelined Scan Enable.

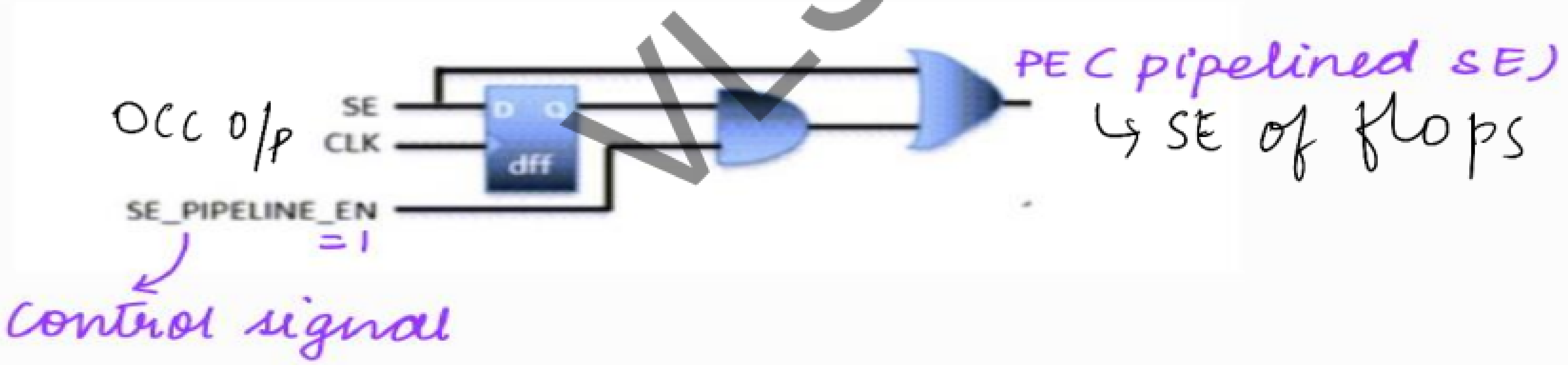
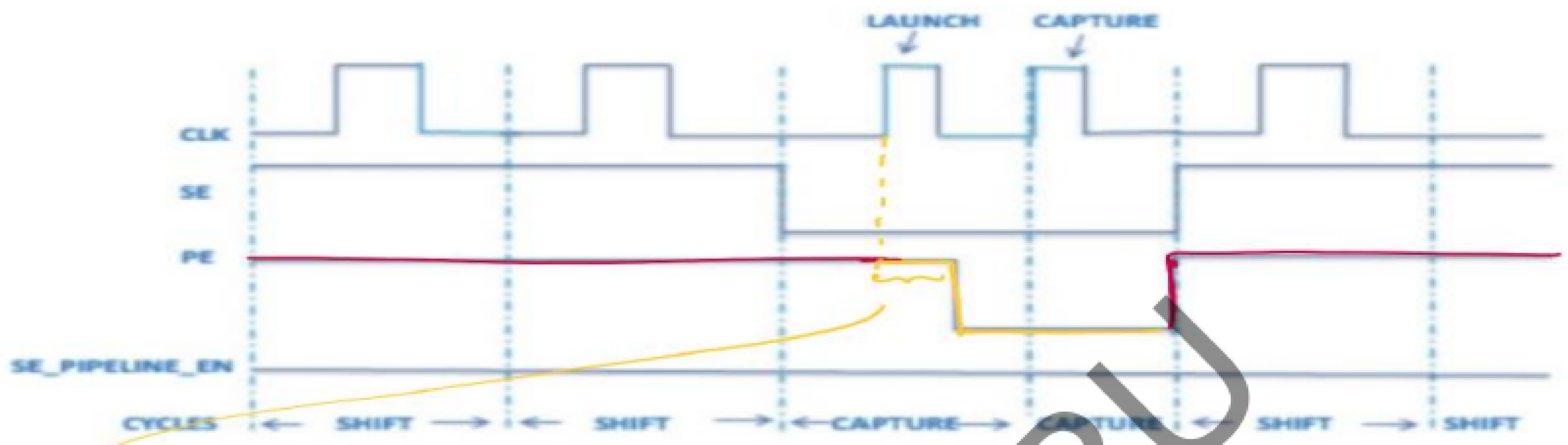
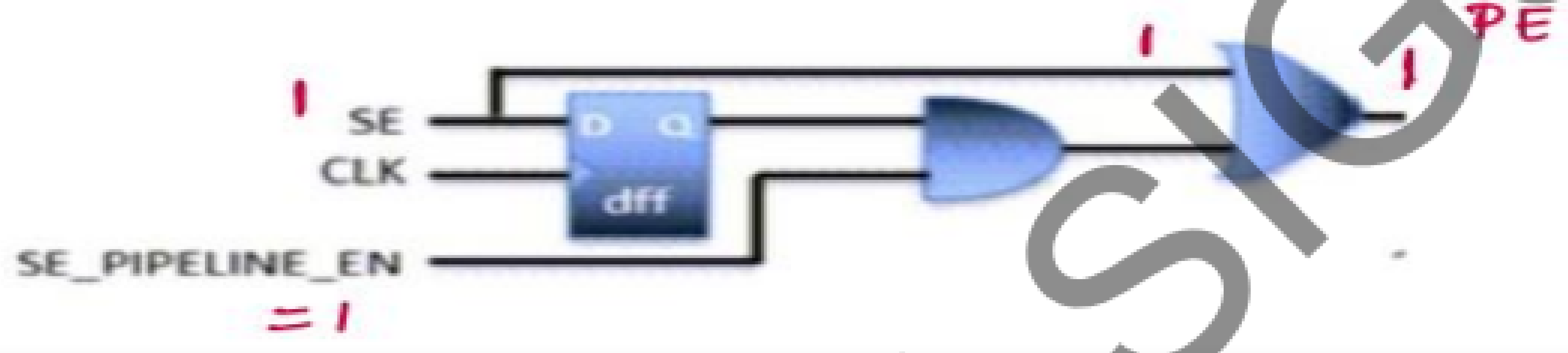


Figure 2. Timing With Scan Enable Pipelining



small delay is because of clk-q delay of the ffs & gate delay.

When SE = 1,



When SE = 0